

Exam 2

1:10 to 3:05 PM

NAME: _____

Question 1 True, False, or Uncertain (20 points). Answer true, false, or uncertain for the following statements and justify your answer with a brief explanation.

- 1) (5 points) The introduction of expectations in the goods market model makes the IS curve flatter, although it is still downward sloping.

- 2) (5 points) In the medium-run open economy, if the nominal exchange rate is fixed, so is the real exchange rate.

- 3) (5 points) The multiplier is larger in an open economy than the one in a closed economy.

- 4) (5 points) An inverted yield curve is a precursor of recession.

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1) (10 points) What is Marshall-Lerner condition? Describe or derive Marshall-Lerner condition briefly.

- 2) (15 points) What is the most memorable topic (or economist) among the ones you have learned in this course? What did you learn from this course? Please write down several constructive feedback to improve the course.

Question 4 (35 points). Consider an open economy with two countries: domestic, and foreign. Each country is characterized by the following behavioral equations:

Domestic Country

$$C = 50 + 0.5(Y - T)$$

$$I = 80 + 0.15Y - 600i$$

$$G = 30$$

$$T = 60$$

$$X = 0.3Y^*$$

$$IM = 0.25Y + 10\frac{EP}{P^*}$$

$$i_t = \bar{i} = 0.03$$

Foreign Country

$$C^* = 50 + 0.5(Y^* - T^*)$$

$$I^* = 70 + 0.3Y^* - 600i^*$$

$$G^* = 30$$

$$T^* = 60$$

$$X^* = 0.25Y + 10\frac{EP}{P^*}$$

$$IM^* = 0.3Y^*$$

$$i_t^* = \bar{i}^* = 0.1$$

Assume that $E_t = 1,000$, $P = 1$, and $P^* = 1,000$.

- 1) (3 points) You will see one of the economies larger than the other. Which economy do you expect is larger? Explain. (*Hint: Compare the propensity to import of two economies and guess how it is related to the scale of the economy.*)

- 2) (4 points) Suppose that the (uncovered) interest parity condition holds in the short-run equilibrium. What is the interest parity condition? Compute the expected future nominal exchange rate E_{t+1}^e by using the approximated interest parity condition. Is the nominal exchange rate E_t expected to appreciate or depreciate?

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- 3) (8 points) Solve for domestic equilibrium output Y , domestic net export NX , foreign equilibrium output Y^* , and foreign net export NX^* . Draw the equilibrium of domestic economy using an $IS - LM - UIP$ diagram.

- 4) (8 points) Now suppose that the domestic government increases G to $G' = 48$. Assuming that the foreign government does not change G^* , solve for new domestic equilibrium output Y_1 , domestic net export NX_1 , foreign equilibrium output Y_1^* , and foreign net export NX_1^* . Draw the effects of domestic expansionary fiscal policy on domestic economy using an $IS - LM - UIP$ diagram.

Now return to the initial situation in which $G = 30$.

- 5) (4 points) Suppose that the foreign country increases the foreign policy rate to $\bar{i}^{*'} = 0.125$ in order to reduce price level to $P^{*'} = 250$. To maintain $E_t = 1,000$ and previous expected future nominal exchange rate E_{t+1}^e , in what direction and how much must the domestic central bank change $\bar{i}$? Use the approximated interest parity condition.

- 6) (8 points) Solve for new domestic equilibrium output Y_2 , domestic net export NX_2 , foreign equilibrium output Y_2^* , and foreign net export NX_2^* . Draw the effects of domestic and foreign monetary policy on domestic economy using an $IS - LM - UIP$ diagram. Assume that domestic price level maintains at $P = 1$.

Scratch Paper

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